

Module 5, Read 1

Simple linear regression: building, assessing and using

Module 5, Read 1: Simple linear regression

Project: [Project 5, Looking Ahead](#) · Preview: [Read 0](#) · Prerequisite: [Module 3 scatterplots & correlation](#) · Next: [Quiz A](#),

This reading is your self-paced textbook for Module 5. In [Read 0](#) you reviewed how to read a scatterplot and summarize a linear trend with Pearson r . Here you will work through three stages: Build a fitted line, Assess whether it is trustworthy and how well it predicts, and Use it to test the relationship and attach honest intervals to your predictions, the complete conceptual toolkit for Project 5.

Learning outcomes

Work through the three sections below, taking notes and using the Check yourself box at the end of each before [Quiz A](#).

After this reading you should be able to:

Build:

- Motivate the least-squares line and know that the slope b_1 and intercept b_0 can be by hand (and recognize the matrix form).
- Fit a simple linear regression with `lm()`, interpret slope and intercept in context, write the prediction equation, and make a point prediction by hand and from the R output.

Assess:

- State the conditions for trusting regression / `cor.test` inference: independent observations, linear form, and approximately normal residuals with constant spread.
- Interpret R^2 and MSE in context for a given dataset.
- Explain why R^2 and MSE are optimistic on the training data and should be judged on a test set or by using cross-validation.

Use:

- Test whether the slope β_1 (equivalently, the correlation ρ) differs from zero using `summary(lm())` and `cor.test()`, and explain why these give the same p-value in simple regression.
- Interpret a confidence interval for the slope, a confidence interval for the mean response at x_0 , and a prediction interval for an individual at x_0 , and explain why the PI is wider.

Optional:

- (*Optional*) Recognize simple logistic regression for a binary response and the idea of multiple regression.

1. Build the model

Goal: turn a roughly linear scatterplot into a fitted equation you can read, interpret, and plug numbers into.

From scatterplot to a fitted line

Every simple linear regression has two variables in fixed roles:

- The response y (also called the outcome or dependent variable) is the thing you want to predict.
- The predictor x (also called the explanatory or independent variable) is the thing you predict it from.

When a scatterplot of the response y against the predictor x looks roughly linear, we summarize the trend with a straight line:

$$\hat{y} = b_0 + b_1 x$$

where b_1 is the slope (predicted change in y for a one-unit increase in x) and b_0 is the intercept (predicted y when $x = 0$). The hat on $\hat{y}$ signals a prediction, not an observed value. For example, if y is the height of my sunflower, then $\hat{y}$ is the *predicted* height of my sunflower.

i Prediction is not the same as estimation

These sound alike but are different goals, and statistics keeps them separate:

- Estimation pins down a fixed unknown quantity for a whole group: a parameter such as the slope β_1 , or the mean response (the average y) at a given x . An estimate is reported with a confidence interval.
- Prediction guesses the value of one new individual observation, a single future y . A prediction is reported with a prediction interval.

Why “least squares”?

For any candidate line, each point has a residual, the vertical miss between what we observed and what the line predicts:

$$e_i = y_i - \hat{y}_i.$$

A line that fits well makes these residuals small. The standard choice is to minimize the sum of squared residuals $\sum e_i^2$. Squaring keeps the positive and negative misses from cancelling, penalizes big misses more, and yields “nice” formulas for people without computers. The minimizing line is the **least-squares regression** line.

The picture below ties the pieces together: the data, the fitted line, the residuals it leaves behind, and where the intercept b_0 and slope b_1 live.

Optional activity: [visualize best-fit line choices](#), drag a line and watch the squared error change.

Notation: the population truth vs your sample

Statistics keeps two layers separate, and the symbols tell you which layer you are on.

- Greek letters are the unknown population truths we wish we knew. The theoretical model for the whole population is

$$Y = \beta_0 + \beta_1 X + \varepsilon,$$

where β_0, β_1 are the true intercept and slope, ρ is the true correlation, and ε (epsilon) is the random error: how far an individual falls from the true line.

- Latin letters are what we actually compute from one sample to estimate those truths. Fitting a line gives

$$\hat{y} = b_0 + b_1 x,$$

where b_0, b_1 estimate β_0, β_1 , the sample correlation r estimates ρ , and the residual $e_i = y_i - \hat{y}_i$ is our observed stand-in for the unseen error ε_i .

- Capitals vs lowercase mark random variable vs observed value. X and Y are random variables (the population-level, “could be anything” quantities); x_i and y_i are the specific numbers you actually recorded for case i .
- A hat means “estimated from data.” So $\hat{y}$ is a predicted value, and you will sometimes see $\hat{\beta}_1$ written for what we call b_1 .

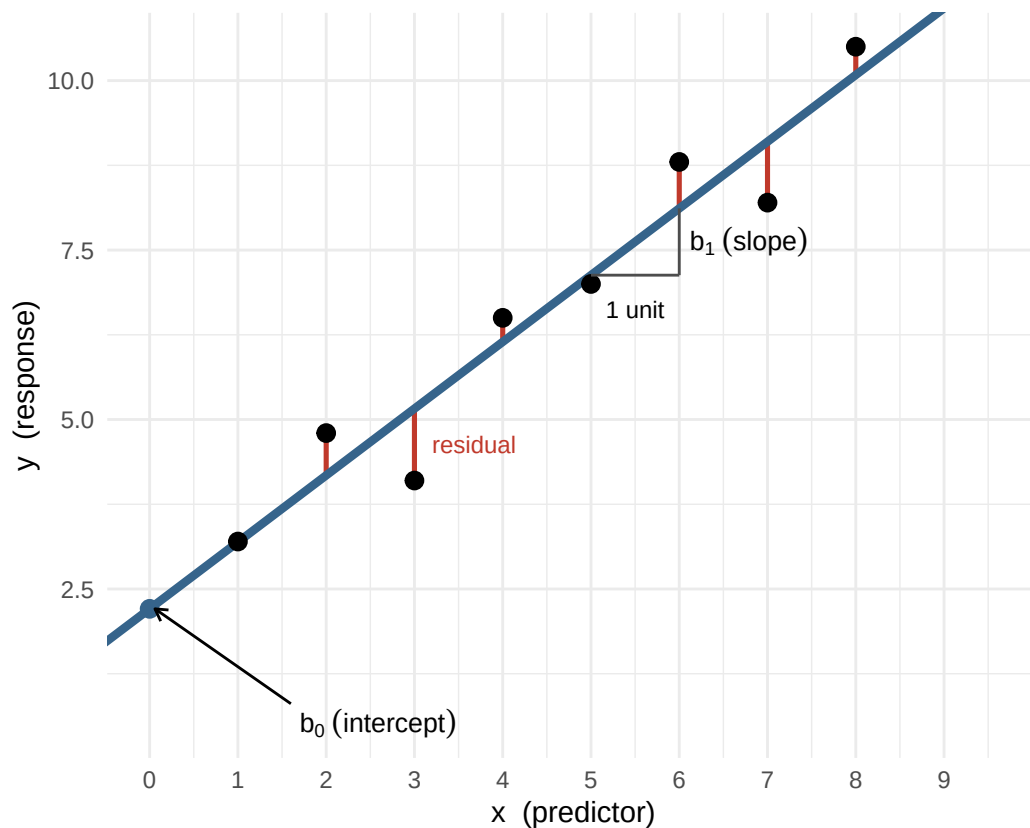


Figure 1: Anatomy of the least-squares line. Black dots are the observed data. The thick blue line is the fitted line, $\hat{y} = b_0 + b_1x$. Each red segment is a residual, the vertical gap between a point and the line ($e_i = y_i - \hat{y}_i$); least squares chooses the one line that makes the sum of the squared red lengths as small as possible. The intercept b_0 is the height of the line at $x = 0$; the slope b_1 is the rise in $\hat{y}$ for a one-unit run in x .

Quantity	Population (truth)	Sample (your data)
Intercept	β_0	b_0
Slope	β_1	b_1
Correlation	ρ	r
Error / residual	ε_i	e_i
Variable	X, Y (random variables)	x_i, y_i (observed values)

The whole point of the rest of this reading: use the sample quantities (b_0, b_1, r, e_i) to learn about the population quantities (β_0, β_1, ρ) we cannot see directly.

The by-hand formulas

Minimizing the total squared residual for a dataset over every possible line gives “nice” formulas for the slope and intercept (the box below shows where they come from):

$$b_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = r \frac{s_y}{s_x}, \quad b_0 = \bar{y} - b_1 \bar{x}.$$

Two things worth noticing:

- r is Pearson’s correlation and s_x and s_y are the standard deviations (spreads) of the variables so multiplying by their ratio puts the right units back on to correlation, giving a measure of the steepness of the linear form rather than the strength without units.
- The slope is just the correlation rescaled by the spreads s_y/s_x . So b_1 and r always share the same sign, a positive correlation means an upward line.
- Because $b_0 = \bar{y} - b_1 \bar{x}$, the line always passes through the point of averages $(\bar{x}, \bar{y})$.

i Optional, where these formulas come from (no calculus)

The only fact we need is from algebra: an upward-opening parabola $f(t) = At^2 + Bt + C$ with $A > 0$ has its lowest point (its vertex) at $t = -\frac{B}{2A}$.

Step 1, the intercept. Hold the slope b_1 fixed and view the total squared residual as a function of b_0 only. Each residual is $y_i - b_1 x_i - b_0$, and squaring and adding gives

$$\text{SSE}(b_0) = n b_0^2 - 2 \left(\sum (y_i - b_1 x_i) \right) b_0 + (\text{a constant}),$$

a parabola in b_0 with $A = n$. Its vertex sits at $b_0 = \frac{\sum (y_i - b_1 x_i)}{n} = \bar{y} - b_1 \bar{x}$, so the best line always passes through $(\bar{x}, \bar{y})$.

Step 2, the slope. Substitute that intercept, which rewrites each residual as $(y_i - \bar{y}) - b_1(x_i - \bar{x})$. As a function of b_1 only,

$$\text{SSE}(b_1) = \left(\sum (x_i - \bar{x})^2 \right) b_1^2 - 2 \left(\sum (x_i - \bar{x})(y_i - \bar{y}) \right) b_1 + (\text{a constant}),$$

a parabola in b_1 with $A = \sum (x_i - \bar{x})^2$. Its vertex sits at

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2},$$

exactly the slope formula above. By minimizing the sum of *squared* residuals, we avoid Calculus and derivatives and just use “the bottom of a parabola sits at $-B/(2A)$.”

i Optional aside, the matrix form

Every linear model can be written $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, where $\mathbf{X}$ stacks a column of 1s (for the intercept) next to the predictor column. The least-squares solution is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

This single expression returns b_0 and b_1 for simple regression, and, with more predictor columns, all the coefficients of multiple regression (the STAT 333 preview at the end). You never have to compute it by hand; `lm()` does it for you. It is shown only so the formula looks familiar later.

See it once: by hand and with R

Unless you get stuck on a desert island, you probably won't compute the slope and intercept by hand. Their formulas can be computed by hand but R does it quickly and accurately. Here is a tiny dataset so you can see what the calculations would look like and to match them up with the R output. Let's take five points and build the pieces the slope formula needs:

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
1	3	-2	-3	6	4
2	5	-1	-1	1	1
3	4	0	-2	0	0
4	8	1	2	2	1
5	10	2	4	8	4

Here $\bar{x} = 3$ and $\bar{y} = 6$. Listing every term in the formula, the slope is the sum of the fifth column over the sum of the sixth:

$$b_1 = \frac{6 + 1 + 0 + 2 + 8}{4 + 1 + 0 + 1 + 4} = \frac{17}{10} = 1.7,$$

and the intercept follows from the point of averages:

$$b_0 = \bar{y} - b_1 \bar{x} = 6 - (1.7)(3) = 6 - 5.1 = 0.9.$$

So the fitted line is $\hat{y} = 0.9 + 1.7x$. Now fit the very same five points with R:

```
demo <- data.frame(x = c(1, 2, 3, 4, 5),  
                   y = c(3, 5, 4, 8, 10))  
lm(y ~ x, data = demo)
```

Call:

```
lm(formula = y ~ x, data = demo)
```

Coefficients:

```
(Intercept)      x  
      0.9      1.7
```

Find your two hand numbers in the output: the value printed under (Intercept) is the intercept $b_0 = 0.9$, and the value under x is the slope $b_1 = 1.7$. Same answer, two routes.

The running example: study hours and exam score

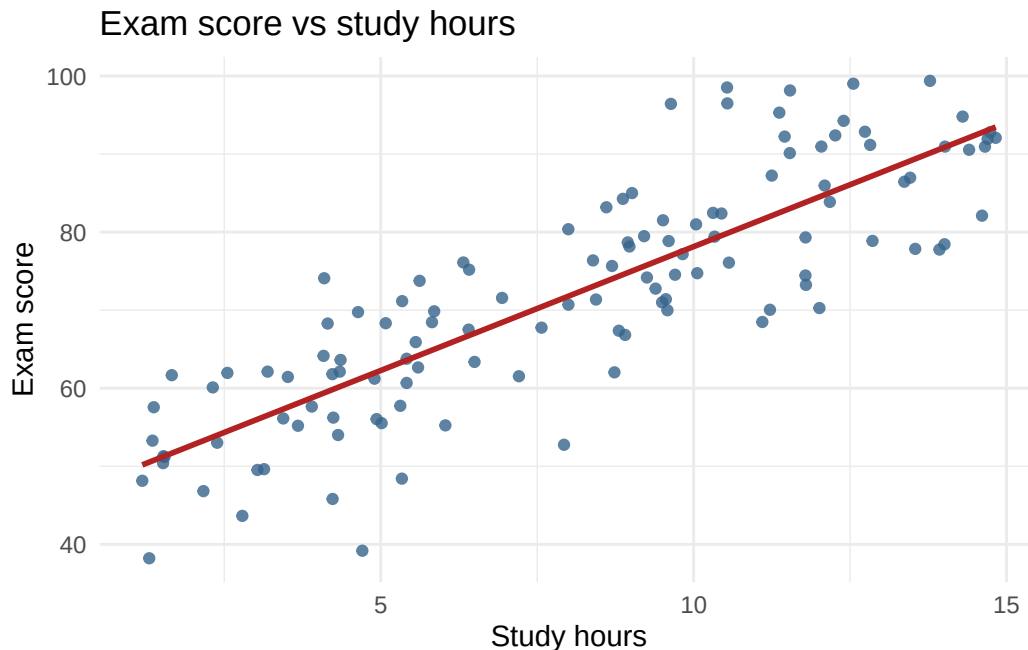
In the rest of this reading we use a larger, more realistic dataset of 120 students, study hours and exam score:

```
library(tidyverse)

set.seed(3401)
n <- 120
exam_dat <- tibble(
  study_hours = runif(n, 1, 15),
  exam_score = 45 + 3.4 * study_hours + rnorm(n, 0, 8)
)
```

First confirm the form is linear (so a line is appropriate model):

```
ggplot(exam_dat, aes(x = study_hours, y = exam_score)) +
  geom_point(alpha = 0.8, color = "steelblue4") +
  geom_smooth(method = "lm", se = FALSE, color = "firebrick") +
  labs(title = "Exam score vs study hours",
       x = "Study hours", y = "Exam score") +
  theme_minimal()
```



With a linear trend confirmed, let R compute the same sums for all 120 points and return the slope and intercept:

```
fit_exam <- lm(exam_score ~ study_hours, data = exam_dat)
fit_exam
```

Call:

```
lm(formula = exam_score ~ study_hours, data = exam_dat)
```

Coefficients:

```
(Intercept)  study_hours
    46.400      3.174
```

Interpret the slope and intercept

- Slope: each additional hour of study is associated with a predicted increase of about 3.17 points.
- Intercept: the predicted score at 0 study hours is about 46.4 points (an extrapolation outside the data, interpret

with care).

Write and use the equation

Read b_0 and b_1 from the R output and write the prediction equation, $\hat{y} = b_0 + b_1x$. A point prediction plugs an x value into the equation, either by hand or with `predict()`:

```
predict(fit_exam, newdata = tibble(study_hours = 10))
```

```
1  
78.14102
```

i The best single guess is the mean

There is a clean theory behind this one number. If you must commit to a single value as your guess for a quantity and you are judged by squared error (the same yardstick least squares uses), the guess that minimizes your average squared miss is the mean: $E[(Y - c)^2] = \text{Var}(Y) + (E[Y] - c)^2$ is smallest exactly at $c = E[Y]$. With a predictor x in hand, that becomes the mean of y at that x , the conditional mean $E[Y | x]$, which is exactly what the regression line estimates. So the fitted value $\hat{y}$ does double duty: it is our estimate of the average y at x_0 and, at the same time, our single best prediction for one new individual at x_0 . That is why the mean-response interval and the prediction interval in Section 3 are centered on the same number, they differ only in width. (Switch the yardstick to absolute error and the best single guess becomes the median instead, which is part of why least squares and the mean go together.)

Check yourself, Build

1. Of all possible straight lines through a scatterplot, which single quantity does the least-squares line make as small as possible?
2. An `lm()` fit of exam score on study hours prints `(Intercept)` 57.6 and `study_hours` 1.8. Write the prediction equation $\hat{y} = b_0 + b_1x$ and interpret the slope in context.
3. Using that equation, predict y at $x = 10$, and define the residual for an observed $y = 90$ at $x = 10$.

Answers, Build

1. The sum of the squared residuals $\sum (y_i - \hat{y}_i)^2$. No other line gives a smaller total squared vertical miss; this is the criterion `lm()` solves.
2. Read directly from the output: $\hat{y} = 57.6 + 1.8x$. Slope: each additional study hour is associated with a predicted increase of about 1.8 exam-score points.
3. $\hat{y} = 57.6 + 1.8(10) = 75.6$. Residual = $y - \hat{y} = 90 - 75.6 = 14.4$ (observed 14.4 above the line).

2. Assess the model

Goal: decide whether the fitted line is trustworthy and how accurate its predictions are. Do this before you quote any p-value or interval in Section 3.

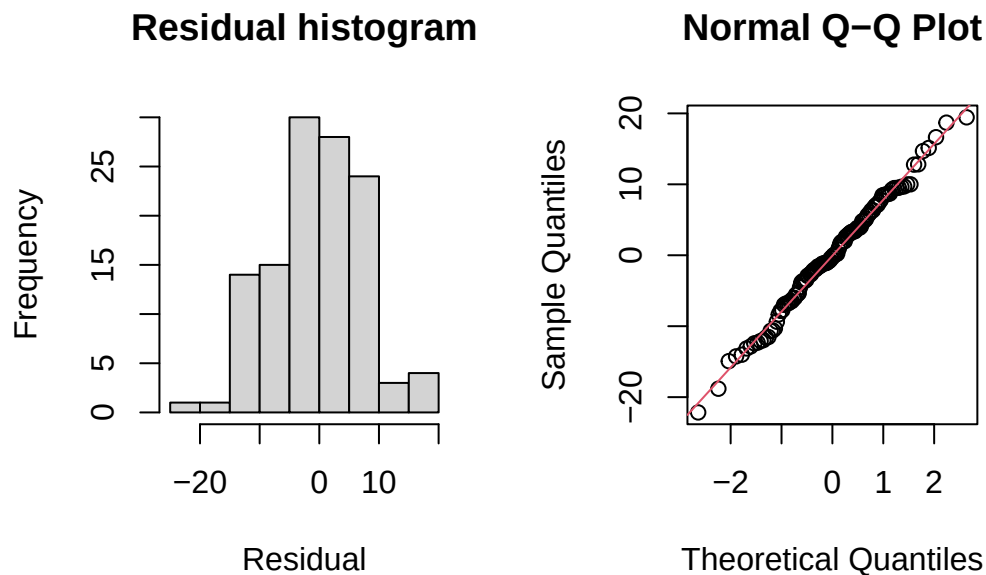
Conditions for trusting inference

The slope/correlation test and every interval in Section 3 rely on conditions. Check them before quoting a p-value or interval built on the model:

- Linear form. The scatterplot trend is straight, not curved. (A p-value for a line fitted to a curve is meaningless.)
- Independent observations. Each row is a separate unit; no repeated measures on the same subject, no time-ordered dependence, no matched pairs.
- Approximately normal residuals with constant spread. The residuals scatter symmetrically around 0 with roughly

equal variance across x (no funnel shape). This is what justifies the t -distribution behind the p-value and intervals. Quick visual checks of the residuals from the exam model, where the conditions hold:

```
par(mfrow = c(1, 2))
hist(residuals(fit_exam), main = "Residual histogram", xlab = "Residual")
qqnorm(residuals(fit_exam)); qqline(residuals(fit_exam), col = 2)
```



```
par(mfrow = c(1, 1))
```

A roughly bell-shaped histogram and points hugging the Q-Q line support the normality condition. A clear funnel in a residuals-vs- x plot would signal non-constant spread and unreliable inference.

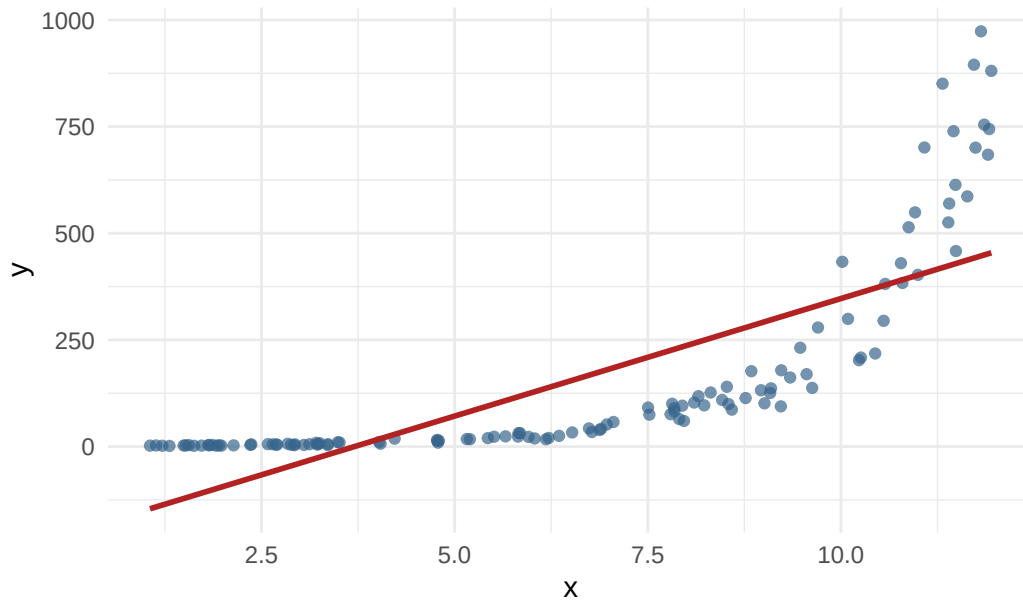
What it looks like when a condition fails

Diagnostics earn their keep when something is wrong. Here a straight line is forced onto a clearly curved relationship:

```
set.seed(3405)
curved <- tibble(
  x = runif(120, 1, 12),
  y = exp(0.55 * x + rnorm(120, 0, 0.25))
)

ggplot(curved, aes(x, y)) +
  geom_point(alpha = 0.7, color = "steelblue4") +
  geom_smooth(method = "lm", se = FALSE, color = "firebrick") +
  labs(title = "A line forced onto a curved relationship",
       x = "x", y = "y") +
  theme_minimal()
```

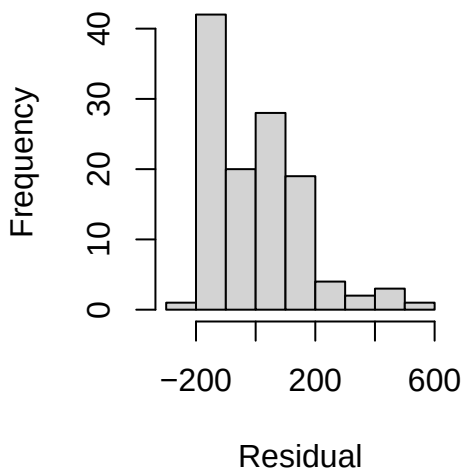
A line forced onto a curved relationship



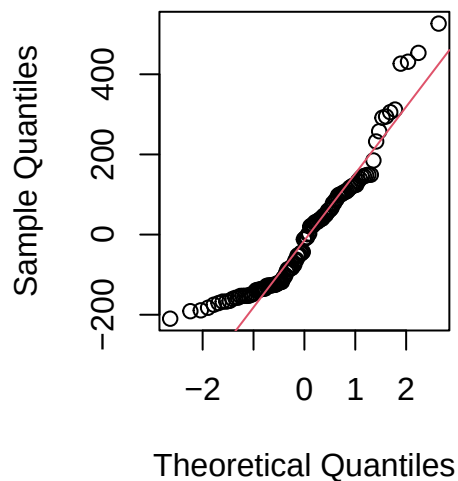
The line cannot follow the curve: it runs above the points through the middle and far below them at the right. That structure leaks straight into the residuals:

```
fit_curved <- lm(y ~ x, data = curved)
par(mfrow = c(1, 2))
hist(residuals(fit_curved), main = "Residual histogram", xlab = "Residual")
qqnorm(residuals(fit_curved)); qqline(residuals(fit_curved), col = 2)
```

Residual histogram



Normal Q-Q Plot



```
par(mfrow = c(1, 1))
```

Now the histogram is sharply right-skewed instead of bell-shaped, and the Q-Q points swing far off the line in a pronounced curve. Both say the normal-residual condition fails, so the slope p-value and the intervals here are not trustworthy. The fix is not to quote the p-value anyway; it is to go back and model the curve (transform y or add a

curved term, a topic for the second course). For Project 5, the practical takeaway is simpler: confirm your scatterplot looks straight *before* you fit a line.

How good is the prediction? Judge it on new data

Two common summaries of fit:

- Mean squared error (MSE): $\frac{1}{n} \sum (y_i - \hat{y}_i)^2$, the average squared prediction miss (in squared y -units).
- R^2 : the proportion of variation in y explained by the line, $R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$, between 0 and 1.

The optimism trap

If you compute MSE or R^2 on the same data you used to fit the line, the numbers are optimistically biased, the line was tuned to those exact points, so it looks better than it will on anything new. Training error always understates real prediction error. To estimate how the model performs on *new* cases, evaluate it on data it has not seen.

Train/test split

Hold out a random slice of the data, fit on the rest, and score on the held-out part:

```
set.seed(3402)
idx_train <- sample(seq_len(nrow(exam_dat)), size = 0.8 * nrow(exam_dat))
exam_train <- exam_dat[idx_train, ]
exam_test  <- exam_dat[-idx_train, ]

fit_train <- lm(exam_score ~ study_hours, data = exam_train)

mse <- function(y, yhat) mean((y - yhat)^2)
r2  <- function(y, yhat) 1 - sum((y - yhat)^2) / sum((y - mean(y))^2)

train_pred <- predict(fit_train, newdata = exam_train)
test_pred  <- predict(fit_train, newdata = exam_test)

tibble(
  set = c("train", "test"),
  MSE = c(mse(exam_train$exam_score, train_pred), mse(exam_test$exam_score, test_pred)),
  R2  = c(r2(exam_train$exam_score, train_pred),  r2(exam_test$exam_score, test_pred))
)

# A tibble: 2 x 3
  set      MSE    R2
  <chr> <dbl> <dbl>
1 train  55.4 0.733
2 test   91.8 0.623
```

Report the test MSE and test R^2 as your honest performance estimate.

Cross-validation: use every point as a test point

A single split wastes 20% of the data and is noisy. k -fold cross-validation splits the data into k parts, and in turn holds out each part as the test set while fitting on the other $k - 1$. Averaging the k test MSEs gives a stable estimate that uses all the data:

```
set.seed(3403)
k <- 5
```

```

folds <- sample(rep(1:k, length.out = nrow(exam_dat)))

cv_mse <- map_dbl(1:k, function(f) {
  tr <- exam_dat[folds != f, ]
  te <- exam_dat[folds == f, ]
  m <- lm(exam_score ~ study_hours, data = tr)
  mse(te$exam_score, predict(m, newdata = te))
})

c(per_fold = round(cv_mse, 1), cv_MSE = round(mean(cv_mse), 1))

```

per_fold1	per_fold2	per_fold3	per_fold4	per_fold5	cv_MSE
55.3	82.6	54.6	66.6	63.8	64.6

The averaged CV MSE is the number to trust when comparing candidate predictors for your project.

Two worked examples on real data

Numbers mean little in the abstract, so here are two regressions on datasets you met in earlier modules. For each we fit on 80% of the rows and read slope, intercept, R^2 , and MSE off the held-out 20%.

Example 1, Palmer penguins: predict body mass (g) from flipper length (mm).

```

if (!requireNamespace("palmerpenguins", quietly = TRUE)) {
  install.packages("palmerpenguins", repos = "https://cloud.r-project.org")
}
library(palmerpenguins)

peng <- penguins |> drop_na(flipper_length_mm, body_mass_g)

set.seed(511)
ip <- sample(seq_len(nrow(peng)), size = 0.8 * nrow(peng))
fit_peng <- lm(body_mass_g ~ flipper_length_mm, data = peng[ip, ])
peng_test <- peng[-ip, ]
peng_yhat <- predict(fit_peng, newdata = peng_test)
peng_mse <- mse(peng_test$body_mass_g, peng_yhat)
peng_r2 <- r2(peng_test$body_mass_g, peng_yhat)

coef(fit_peng)

```

(Intercept)	flipper_length_mm
-5615.09262	48.87733

```

c(test_MSE = peng_mse, test_RMSE = sqrt(peng_mse), test_R2 = peng_r2)

```

test_MSE	test_RMSE	test_R2
1.267859e+05	3.560701e+02	8.311173e-01

Reading these in context:

- Slope (about 48.9 g/mm): each additional millimeter of flipper length is associated with a predicted increase of roughly 48.9 grams in body mass.
- Intercept (about -5,615 g): the predicted body mass at a flipper length of 0 mm. No penguin has a 0 mm flipper, so this is a meaningless extrapolation; the intercept just anchors the height of the line.
- Test R^2 (about 0.83): flipper length alone explains roughly 83% of the variation in body mass among the held-out

penguins.

- Test RMSE (about 356 g): a prediction for a new penguin typically misses its actual mass by that many grams. (RMSE is the square root of MSE, back in grams, which is easier to read than MSE in grams-squared.)

Example 2, diamonds: predict price (\$) from carat.

```
set.seed(512)
id <- sample(seq_len(nrow(diamonds)), size = 0.8 * nrow(diamonds))
fit_dia <- lm(price ~ carat, data = diamonds[id, ])
dia_test <- diamonds[-id, ]
dia_yhat <- predict(fit_dia, newdata = dia_test)
dia_mse <- mse(dia_test$price, dia_yhat)
dia_r2 <- r2(dia_test$price, dia_yhat)

coef(fit_dia)
```

```
(Intercept)      carat
-2274.604      7786.598
```

```
c(test_MSE = dia_mse, test_RMSE = sqrt(dia_mse), test_R2 = dia_r2)
```

```
      test_MSE      test_RMSE      test_R2
2.358774e+06 1.535830e+03 8.492818e-01
```

Reading these in context:

- Slope (about \$7,787 per carat): each additional carat is associated with a predicted increase of about that many dollars in price.
- Intercept (about \$-2,275): the predicted price of a 0-carat diamond, which comes out negative and physically impossible, another reminder that an intercept is often just a mathematical anchor, not a real prediction.
- Test R^2 (about 0.85): carat alone explains roughly 85% of the variation in price among held-out diamonds, a strong single-predictor fit.
- Test RMSE (about \$1,536): a typical price prediction is off by about that many dollars, large in absolute terms because prices are large and fan out at high carat (a hint the constant-spread condition is strained here, something a second course would address).

Check yourself, Assess

1. List the three conditions you should check before trusting the slope p-value.
2. A classmate reports $R^2 = 0.95$ computed on the same data used to fit the model and claims excellent prediction. What is the flaw, and what should they report instead?
3. For the penguin model above, the held-out fit has test $R^2 \approx 0.76$ and test RMSE ≈ 390 g. Put each of these two numbers into one sentence of plain-language meaning.

Answers, Assess

1. Linear form, independent observations, and approximately normal residuals with constant spread.
2. R^2 on the training data is optimistically biased because the model was fit to those points. They should report test-set (or cross-validated) MSE / R^2 .
3. Test $R^2 \approx 0.76$: flipper length explains about 76% of the variation in body mass among new penguins the model never saw. Test RMSE ≈ 390 g: a typical prediction for a new penguin misses its actual mass by about 390 grams.

3. Use the model

Goal: answer the research question (is there a relationship?) and make predictions with honest intervals. With the conditions in Section 2 met, R supplies the test and three intervals.

Is the relationship real? Testing the slope and the correlation

A fitted line always has *some* slope, even for a cloud of noise. A hypothesis asks whether the *population* slope is distinguishable from zero assuming the simple linear model is the underlying truth.

- Null: $H_0 : \beta_1 = 0$ (no linear relationship; equivalently $H_0 : \rho = 0$).
- Alternative: $H_a : \beta_1 \neq 0$ (a linear relationship exists).

`summary(lm())` reports the t-statistic and p-value for the slope:

```
summary(fit_exam)
```

Call:

```
lm(formula = exam_score ~ study_hours, data = exam_dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-22.1569	-5.3780	-0.2521	5.2489	19.4441

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	46.4005	1.6648	27.87	<2e-16 ***
study_hours	3.1741	0.1856	17.10	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.961 on 118 degrees of freedom

Multiple R-squared: 0.7126, Adjusted R-squared: 0.7101

F-statistic: 292.5 on 1 and 118 DF, p-value: < 2.2e-16

Read the `study_hours` row: the Estimate is b_1 , Std. Error is its standard error, `t value` = b_1/SE , and `Pr(>|t|)` is the two-sided p-value for $H_0 : \beta_1 = 0$.

The equivalent correlation test asks $H_0 : \rho = 0$:

```
cor.test(exam_dat$study_hours, exam_dat$exam_score)
```

Pearson's product-moment correlation

data: exam_dat\$study_hours and exam_dat\$exam_score

t = 17.103, df = 118, p-value < 2.2e-16

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

0.7834263 0.8888828

sample estimates:

cor

0.8441325

i Same question, same p-value

In simple linear regression the slope test and the correlation test are algebraically the same test, they give the identical p-value. Testing “is the slope zero?” *is* testing “is the correlation zero?” because $b_1 = r(s_y/s_x)$. Use whichever output is handy; `cor.test()` also returns a confidence interval for ρ itself.

Confidence interval for the slope β_1

A point prediction is a single number; an interval communicates uncertainty. Start with the slope itself:

```
confint(fit_exam, "study_hours", level = 0.95)
```

```
          2.5 %    97.5 %  
study_hours 2.80655 3.541557
```

Interpretation: we are 95% confident that each additional study hour changes the mean exam score by an amount in this interval. If the interval excludes 0, that matches a small slope p-value.

Confidence interval for the mean response at x_0 (estimation)

This is estimation: we pin down the average y , a fixed population quantity. “Among *all* students who study 10 hours, where is the average score?”

```
x0 <- tibble(study_hours = 10)  
predict(fit_exam, newdata = x0, interval = "confidence", level = 0.95)
```

```
          fit      lwr      upr  
1 78.14102 76.53677 79.74526
```

Prediction interval for an individual at x_0 (prediction)

This is prediction: we guess one new individual y , not an average. “For one future student who studies 10 hours, where will *their* score land?”

```
predict(fit_exam, newdata = x0, interval = "prediction", level = 0.95)
```

```
          fit      lwr      upr  
1 78.14102 62.29502 93.98702
```

i Why the prediction interval is always wider

The mean-response CI captures only uncertainty about where the line sits. The prediction interval adds the individual-to-individual scatter around the line on top of that. Two sources of uncertainty instead of one, so the PI is always wider than the CI at the same x_0 . For Project 5 you report a prediction interval, because you are predicting one specific object.

i Bringing it to Project 5

In Project 5 your model is `lm(number ~ total_weight)`: the response y is the number of stones, the predictor x is the total weight. To predict the count in Rosanna’s cup, set $x_0 = 209.27$ g and report `predict(fit, newdata = data.frame(total_weight = 209.27), interval = "prediction")`, the prediction interval, not the mean-response CI, because you are predicting one specific cup.

Check yourself, Use

1. `summary(lm())` gives a slope p-value of 0.002. What do `cor.test()` and this slope test have in common, and what do you conclude?
2. At $x_0 = 10$, the mean-response CI is [76, 80] and the prediction interval is [62, 94]. Explain which is which and why one is wider.

Answers, Use

1. In simple regression they are the same test (identical p-value): testing $\beta_1 = 0$ is testing $\rho = 0$. With $p = 0.002$ we reject H_0 , there is a statistically significant linear relationship.
2. [76, 80] is the mean-response CI (narrower, uncertainty about the average); [62, 94] is the prediction interval (wider, it adds individual-to-individual scatter on top of the mean uncertainty).

Interpretation templates at a glance

Use this as a quick reference for Quiz A. Replace the bracketed parts with the variable names and units from the problem, and read L and U as the lower and upper numbers an interval reports.

Quantity	Where it comes from	One-sentence interpretation template
Slope b_1	<code>coef(fit)</code> , predictor row	Each additional 1 [x-unit] is associated with a predicted change of b_1 [y-units] in [y].
Intercept b_0	<code>coef(fit)</code> , (Intercept) row	The predicted [y] when [x] = 0 is b_0 [y-units] (meaningful only if $x = 0$ is realistic).
Point prediction $\hat{y}$	<code>predict(fit, newdata)</code>	The predicted [y] for [x] = x_0 is $\hat{y}$ [y-units].
Residual	$y - \hat{y}$	The observed [y] sits this many [y-units] above (or below) the line.
R^2 (test set)	<code>r2(test_y, yhat)</code>	About $R^2 \times 100\%$ of the variation in [y] is explained by [x]; judge it on held-out data.
MSE / RMSE (test set)	<code>mse(...)</code> , <code>sqrt(mse(...))</code>	RMSE: a typical prediction misses the actual [y] by about this many [y-units]. $\text{RMSE} = \sqrt{\text{MSE}}$, back in [y]'s units.
CI for the slope	<code>confint(fit, "x")</code>	We are 95% confident each additional 1 [x-unit] changes the mean [y] by between L and U [y-units].
CI for the intercept	<code>confint(fit, "(Intercept)")</code>	We are 95% confident the mean [y] when [x] = 0 is between L and U [y-units].
CI for the mean response	<code>predict(..., interval = "confidence")</code>	We are 95% confident the mean [y] for all units with [x] = x_0 is between L and U [y-units].
Prediction interval	<code>predict(..., interval = "prediction")</code>	We are 95% confident one new individual with [x] = x_0 has [y] between L and U [y-units].

The last four rows all report an interval $[L, U]$; what changes is the quantity it describes. `confint()` gives an interval for a coefficient (slope or intercept), `predict(..., interval = "confidence")` gives the mean response at x_0 , and `predict(..., interval = "prediction")` gives one individual at x_0 (always the widest). On Quiz A you match the R call to the matching template.

Optional: beyond the simple straight line

These sections are not required for Project 5 or Quiz A, they show where the ideas go next.

Simple logistic regression (binary response)

When the response is binary (yes/no, success/failure) rather than numerical, a straight line can predict impossible values below 0 or above 1. Logistic regression instead models the *probability* of a “yes” with an S-shaped curve, fit with `glm(..., family = binomial)`:

```
set.seed(3404)
logit_dat <- tibble(
  hours = runif(150, 0, 15),
  passed = rbinom(150, 1, plogis(-4 + 0.6 * hours))
)
fit_log <- glm(passed ~ hours, data = logit_dat, family = binomial)
coef(fit_log)
```

(Intercept)	hours
-4.925942	0.726906

The coefficients are on the log-odds scale; exponentiating a slope gives an odds ratio (the multiplicative change in odds per one-unit increase in x). The mechanics are a full topic of their own.

A preview of multiple regression (STAT 333)

Real predictions usually use several predictors at once. Multiple regression extends the same least-squares idea to

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + \cdots + b_kx_k,$$

fit with `lm(y ~ x1 + x2 + ...)`, exactly the matrix formula from Section 1 with more columns in $\mathbf{X}$. Choosing predictors, checking conditions, and avoiding overfitting are the heart of STAT 333 (the second statistics course at Cal Poly Humboldt). Module 5 stays with a single predictor so you can see every piece clearly.

When you finish, take [Quiz A](#), then move on to the [demo lab](#).